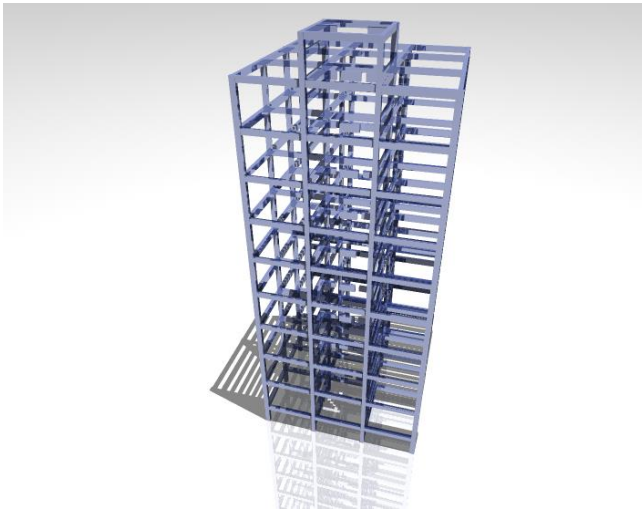




MULTISTOREY BUILDING DESIGN ANALYSIS IN GOPALGANJ CITY

A design analysis conducted with hard work.

ABSTRACT

THIS PROJECT PRESENTS THE SEISMIC ANALYSIS OF A MULTI-STORY RESIDENTIAL BUILDING IN GOPALGANJ CITY USING ETABS 2022, FOLLOWING THE BANGLADESH NATIONAL BUILDING CODE (BNBC) 2017 GUIDELINES. THE STUDY FOCUSES ON MODELING THE STRUCTURE, APPLYING SEISMIC LOADS, AND EVALUATING KEY PARAMETERS SUCH AS BASE SHEAR AND STORY DRIFT. PRELIMINARY COMPARISONS WITH HAND CALCULATIONS WERE CONDUCTED, THOUGH SOME CORRECTIONS, INCLUDING SEISMIC ZONE ADJUSTMENTS, REMAIN INCOMPLETE. THIS WORK PROVIDES A FOUNDATIONAL UNDERSTANDING OF EARTHQUAKE-RESISTANT DESIGN AND HIGHLIGHTS AREAS FOR FURTHER REFINEMENT.

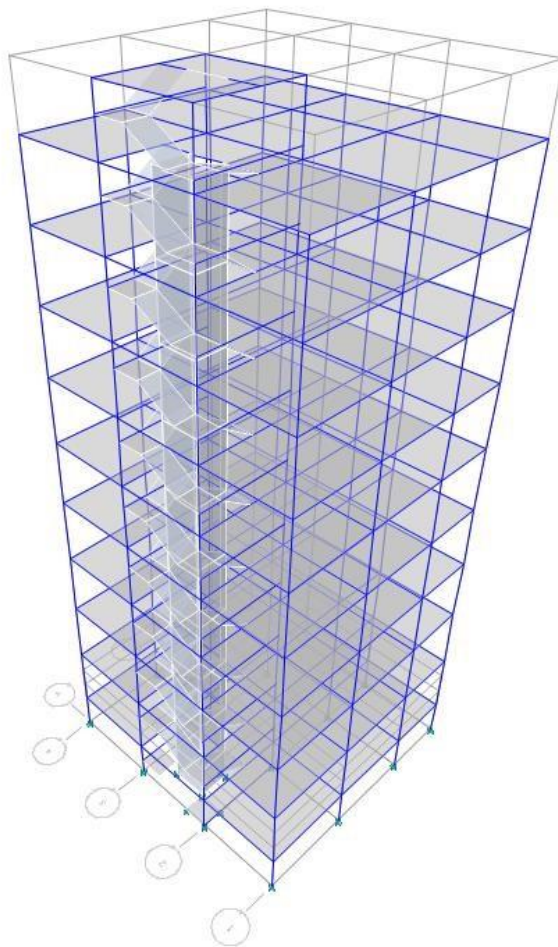
Sadman Ahmed Sabub

ETABS-2022 Orientation

Project- *“Multistorey Building Design Analysis in Gopalganj City”*

The ETABS report for the project:

ETABS[®]



Project Report

Model File: Seismic Analysis of a Multi-storey Building inside Dhaka city, Revision 0

1/16/2025

Table of Contents

1. Structure Data	7
1.1 Story Data	7
1.2 Grid Data	7
1.3 Point Coordinates	7
1.4 Line Connectivity	9
1.5 Area Connectivity	10
1.6 Mass	12
2. Properties	14
2.1 Materials	14
2.2 Frame Sections	14
2.3 Shell Sections	16
2.4 Reinforcement Sizes	16
3. Loads	16
3.1 Load Patterns	16
3.2 Auto Wind Loading	16
3.3 Auto Seismic Loading	19
3.4 Load Cases	1
3.5 Load Combinations	1

List of Tables

Table 1.1 Story Definitions	4
Table 1.2 Grid Definitions - General	4
Table 1.3 Grid Definitions - Grid Lines	4
Table 1.4 Point Bays	4
Table 1.5 Column Bays	6
Table 1.6 Beam Bays	6
Table 1.7 Floor Bays	7
Table 1.8 Wall Bays	8
Table 1.9 Mass Source Definition	9
Table 2.1 Material Properties - General	10
Table 2.2 Frame Section Property Definitions - Summary	10
Table 2.3 Area Section Property Definitions - Summary	11
Table 2.4 Reinforcing Bar Sizes	11
Table 3.1 Load Pattern Definitions	12
Table 3.2 Load Case Definitions - Summary	17
Table 3.3 Load Combination Definitions	17

1 Structure Data

This chapter provides model geometry information, including items such as story levels, point coordinates, and element connectivity.

1.1 Story Data

Table 1.1 - Story Definitions

Tower	Name	Height ft	Master Story	Similar To	Splice Story	Color
T1	Rooftop	8	No	None	No	Red
T1	Story10	10	No	None	No	Red
T1	Story9	10	No	None	No	Red
T1	Story8	10	No	None	No	Red
T1	Story7	10	No	None	No	Red
T1	Story6	10	No	None	No	Red
T1	Story5	10	No	None	No	Red
T1	Story4	10	No	None	No	Red
T1	Story3	10	No	None	No	Red
T1	Story2	10	No	None	No	Red
T1	Story1	6	No	None	No	Magenta

1.2 Grid Data

Table 1.2 - Grid Definitions - General

Tower	Name	Type	Ux ft	Uy ft	Rz deg	Story Range	Bubble Size in	Color
T1	G1	Cartesian	0	0	0	Default	60	Gray6

Table 1.3 - Grid Definitions - Grid Lines

Name	Grid Line Type	ID	Ordinate ft	Bubble Location	Visible
G1	X (Cartesian)	A	0	End	Yes
G1	X (Cartesian)	B	15	End	Yes
G1	X (Cartesian)	C	29	End	Yes
G1	X (Cartesian)	D	39.25	End	Yes
G1	Y (Cartesian)	1	0	Start	Yes
G1	Y (Cartesian)	2	13.6667	Start	Yes
G1	Y (Cartesian)	3	27.5	Start	Yes
G1	Y (Cartesian)	4	41.1667	Start	Yes

1.3 Point Coordinates

Table 1.4 - Point Bays

Label	Is Auto Point	X ft	Y ft	DZBelow ft
1	No	0	0	0
2	No	0	13.6667	0
3	No	0	27.5	0

4	No	0	41.1667	0
5	No	15	0	0
6	No	15	13.6667	0
7	No	15	27.5	0
8	No	15	41.1667	0
9	No	29	0	0

Label	Is Auto Point	X ft	Y ft	DZBelow ft
10	No	29	13.6667	0
11	No	29	27.5	0
12	No	29	41.1667	0
13	No	39.25	0	0
14	No	39.25	13.6667	0
15	No	39.25	27.5	0
16	No	39.25	41.1667	0
17	No	-8	52	0
18	No	3.5	13.6667	0
19	No	3.5	27.5	0
20	No	0	17.6667	0
21	No	0	23.5	0
22	Yes	7.5	20.5834	0
23	No	3.5	17.6667	0
24	No	3.5	23.5	0
25	No	8.5	23.5	0
26	No	8.5	17.6667	0
27	No	8.5	18.6667	0
28	No	8.5	22.5	0
29	No	8.5	27.5	0
30	No	8.5	13.6667	0
31	No	0	17.6667	6.6667
32	No	3.5	13.6667	6.6667
33	No	3.5	27.5	6.6667
34	No	3.5	17.6667	6.6667
35	No	0	13.6667	6.6667
36	No	15	13.6667	6.6667
37	No	3.5	27.5	3.3334
38	No	0	23.5	3.3334
39	No	0	27.5	3.3334
40	No	3.5	23.5	3.3334
41	No	15	27.5	3.3334
42	Yes	19.625	20.5834	0
43	Yes	19.625	20.5834	0
44	Yes	19.625	20.5834	0
45	Yes	19.625	20.5834	0
46	Yes	19.625	20.5834	0
47	Yes	19.625	20.5834	0
48	Yes	19.625	20.5834	0
49	Yes	19.625	20.5834	0
50	Yes	19.625	20.5834	0
51	Yes	7.6265	21.0779	0
52	Yes	19.6808	20.5254	0
53	Yes	20.1669	20.6509	0

54	Yes	20.256	20.5834	0
55	Yes	20.1804	20.5834	0
56	Yes	20.1666	20.6201	0
57	Yes	20.1212	20.543	0
58	Yes	20.114	20.5817	0
59	Yes	20.1432	20.5813	0
60	Yes	20.128	20.5834	0
81	Yes	7.5	20.5834	0
82	Yes	19.625	20.5834	0
83	Yes	19.625	20.5834	0
84	Yes	19.625	20.5834	0
Label	Is Auto Point	X ft	Y ft	DZBelow ft
85	Yes	19.625	20.5834	0
86	Yes	19.625	20.5834	0
87	Yes	19.625	20.5834	0
88	Yes	19.625	20.5834	0
89	Yes	19.625	20.5834	0
90	Yes	19.625	20.5834	0

1.4 Line Connectivity

Table 1.5 - Column Bays

Label	PointBayI	PointBayJ	IEndStory
C1	8	8	Below
C2	16	16	Below
C3	3	3	Below
C4	2	2	Below
C5	11	11	Below
C6	10	10	Below
C7	4	4	Below
C8	12	12	Below
C9	7	7	Below
C10	15	15	Below
C11	6	6	Below
C12	1	1	Below
C13	14	14	Below
C14	9	9	Below
C15	5	5	Below
C16	13	13	Below

Table 1.6 - Beam Bays

Label	PointBayI	PointBayJ
B1	1	2
B2	2	3
B3	3	4
B4	5	6
B5	6	7
B6	7	8
B7	9	10
B8	10	11
B9	11	12

B10	13	14
B11	14	15
B12	15	16
B13	1	5
B14	5	9
B15	9	13
B16	2	6
B17	6	10
B18	10	14
B19	3	7
B20	7	11
B21	11	15
B22	4	8
B23	8	12
B24	12	16
Label	PointBayI	PointBayJ
B25	27	28
B26	25	29
B27	30	26
B28	35	36
B29	39	41

1.5 Area Connectivity

Table 1.7 - Floor Bays

Label	NumPoints	PointNumber	PointBay
F1	4	1	3
F1		2	7
F1		3	8
F1		4	4
F2	4	1	7
F2		2	11
F2		3	12
F2		4	8
F3	4	1	11
F3		2	15
F3		3	16
F3		4	12
F4	4	1	6
F4		2	10
F4		3	11
F4		4	7
F5	4	1	10
F5		2	14
F5		3	15
F5		4	11
F6	4	1	5

F6		2	9
F6		3	10
F6		4	6
F7	4	1	1
F7		2	5
F7		3	6
F7		4	2
F8	4	1	9
F8		2	13
F8		3	14
F8		4	10
F9	4	1	30
F9		2	6
F9		3	7
F9		4	29
F12	4	1	35
F12		2	32
F12		3	34
F12		4	31
F13	4	1	38
F13		2	40
F13		3	37
F13		4	39
F14	4	1	2
F14		2	18
Label	NumPoints	PointNumber	PointBay
F14		3	23
F14		4	20
F15	4	1	30
F15		2	26
F15		3	34
F15		4	32
F17	4	1	2
F17		2	30
F17		3	29
F17		4	3

Name	Is Default	Include Lateral Mass?	Include Vertical Mass?	Lump Mass?	Source Self Mass?	Source Added Mass?	Source Load Patterns?	Move Mass Centroid?	Load Pattern	Multiplier
MsSrc1	Yes	Yes	No	Yes	No	No	Yes	No	Dead	1
MsSrc1									Partition Wall	1

MsSrc1

Floor Finish

1

Table 1.8 - Wall Bays

Label	NumPoints	PointNumber	PointBay	PointStory
W1	4	1	23	Below
W1		2	24	Below
W1		3	24	Same
W1		4	23	Same
W2	4	1	23	Below
W2		2	26	Below
W2		3	26	Same
W2		4	23	Same
W3	4	1	26	Below
W3		2	27	Below
W3		3	27	Same
W3		4	26	Same
W4	4	1	28	Below
W4		2	25	Below
W4		3	25	Same
W4		4	28	Same
W5	4	1	24	Below
W5		2	25	Below
W5		3	25	Same
W5		4	24	Same
W6	4	1	30	Below
W6		2	26	Below
W6		3	34	Same
W6		4	32	Same
W7	4	1	31	Same
W7		2	34	Same
W7		3	40	Same
W7		4	38	Same
W8	4	1	37	Same
W8		2	40	Same
W8		3	25	Same
W8		4	29	Same

1.6 Mass

Table 1.9 - Mass Source Definition

Name	Is Default	Include Lateral Mass?	Include Vertical Mass?	Lump Mass?	Source Self Mass?	Source Added Mass?	Source Load Patterns?	Move Mass Centroid?	Load Pattern	Multiplier
MsSrc1									Live	0.25

Properties

2 Properties

This chapter provides property information for materials, frame sections, shell sections, and links.

2.1 Materials

Table 2.1 - Material Properties - General

Material	Type	SymType	Grade	Color	Notes
4000Psi	Concrete	Isotropic	f'c 4000 psi	Magenta	
A416Gr270	Tendon	Uniaxial	Grade 270	Gray8Dark	
A615Gr60	Rebar	Uniaxial	Grade 60	Yellow	

2.2 Frame Sections

Name	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier	I22 Modifier	Mass Modifier	Weight Modifier
B10X15	1	1	1	1	0.35	0.35	1	1

B10X18	1	1	1	0.35		0.35	1	1
B10X21	1	1	1	1	0.35	0.35	1	1

Table 2.2 - Frame Section Property Definitions - Summary (Part 1 of 3)

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4	As2 in2	As3 in2
B10X15	4000Psi	Concrete Rectangular	Magenta	150	2942.16	2812.5	1250	0	125	125
B10X18	4000Psi	Concrete Rectangular	Yellow	180	3921.41	4860	1500	0	150	150
B10X21	4000Psi	Concrete Rectangular	Gray8Dark	210	4914	7717.5	1750	0	175	175
BEAM Auto Select										
C12X15	4000Psi	Concrete Rectangular	Blue	180	4460.59	3375	2160	0	150	150
C12X18	4000Psi	Concrete Rectangular	Green	216	6100.87	5832	2592	0	180	180
C12X21	4000Psi	Concrete Rectangular	Cyan	252	7789.48	9261	3024	0	210	210
C12X24	4000Psi	Concrete Rectangular	Red	288	9500.62	13824	3456	0	240	240
C12X27	4000Psi	Concrete Rectangular	Magenta	324	11222.85	19683	3888	0	270	270
COLUMN Auto Select										
GB		Auto Select								
GB12X15	4000Psi	Concrete Rectangular	Gray8Dark	180	4460.59	3375	2160	0	150	150
GB12X18	4000Psi	Concrete Rectangular	Blue	216	6100.87	5832	2592	0	180	180

Table 2.2 - Frame Section Property Definitions - Summary (Part 2 of 3)

Name	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3	Z33 in3	Z22 in3	R33 in	R22 in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in	PNA Offset 2 in
B10X15	375	375	250	250	562.5	375	4.3301	2.8868	0	0	0	0
B10X18	540	540	300	300	810	450	5.1962	2.8868	0	0	0	0
B10X21	735	735	350	350	1102.5	525	6.0622	2.8868	0	0	0	0
BEAM												
C12X15	450	450	360	360	675	540	4.3301	3.4641	0	0	0	0
C12X18	648	648	432	432	972	648	5.1962	3.4641	0	0	0	0
C12X21	882	882	504	504	1323	756	6.0622	3.4641	0	0	0	0
C12X24	1152	1152	576	576	1728	864	6.9282	3.4641	0	0	0	0
C12X27	1458	1458	648	648	2187	972	7.7942	3.4641	0	0	0	0
COLUMN												
GB												
GB12X15	450	450	360	360	675	540	4.3301	3.4641	0	0	0	0
GB12X18	648	648	432	432	972	648	5.1962	3.4641	0	0	0	0

Table 2.2 - Frame Section Property Definitions - Summary (Part 3 of 3)

Properties

Name	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier	I22 Modifier	Mass Modifier	Weight Modifier
BEAM								
C12X15	1	1	1	1	0.7	0.7	1	1

C12X18	1	1	1	1	0.7	0.7	1	1
C12X21	1	1	1	1	0.7	0.7	1	1
C12X24	1	1	1	1	0.7	0.7	1	1
C12X27	1	1	1	1	0.7	0.7	1	1
COLUMN								
GB								
GB12X15	1	1	1	1	0.35	0.35	1	1
GB12X18	1	1	1	1	0.35	0.35	1	1

2.3 Shell Sections

Table 2.3 - Area Section Property Definitions - Summary

Name	Type	Element Type	Material	Total Thickness in	Deck Material	Deck Depth in
Slab 5"	Slab	Shell-Thin	4000Psi	5		
Slab 8" Stair	Slab	Shell-Thin	4000Psi	8		
Wall 8"	Wall	Shell-Thin	4000Psi	8		

2.4 Reinforcement Sizes

Table 2.4 - Reinforcing Bar Sizes

Name	Diameter in	Area in2
#6	0.75	0.44

3 Loads

This chapter provides loading information as applied to the model.

3.1 Load Patterns

Table 3.1 - Load Pattern Definitions

Name	Is Auto Load	Type	Self Weight Multiplier	Auto Load
Dead	No	Dead	1	
EQ	No	Seismic	0	User Coefficient
Floor Finish	No	Super Dead	0	
Live	No	Live	0	
Partition Wall	No	Super Dead	0	
WL-X	No	Wind	0	ASCE 7-05
WL-Y	No	Wind	0	ASCE 7-05

3.2 Auto Wind Loading

ASCE 7-05 Auto Wind Load Calculation

This calculation presents the automatically generated lateral wind loads for load pattern WL-X according to ASCE

7-05, as calculated by ETABS.

Exposure Parameters

Exposure From = Diaphragms

Exposure Category = B

Wind Direction = 0 degrees

Basic Wind Speed, V [ASCE 6.5.4]

Windward Coefficient, $C_{p,wind}$ [ASCE

6.5.11.2.1]

Leeward Coefficient, $C_{p,lee}$ [ASCE 6.5.11.2.1]

Wind Case = All Cases

Top Story = Rooftop

Bottom Story = Story1

Include Parapet = Yes, Parapet Height = 1

Factors and Coefficients

Gradient Height, z_g [ASCE Table 6-2]

Emperical Exponent, α [ASCE Table 6-2]

Velocity Pressure Exposure Coefficient, K_z ,

$K_z=2.01(<z>/<z_g>)^{<2>/<\alpha>}$ [ASCE Table 6-3]

$$K_z=2.01(<15>/<z_g>)^{<2>/<\alpha>}$$

Topographical Factor, K_{zt} [ASCE 6.5.7.2]

$$<q_z>=0.00256<K_z><K_{zt}><K_d><V>$$

Velocity Pressure, q_z [ASCE 6.5.10 Eq. 6-15]

Design Wind Pressure, p [ASCE 6.5.12.2.1 Eq. 6-17] $p=qG<C_{p,wind}>+<q_h><(G<C_{p,lee}>)>$

Directionality Factor, K_d [ASCE 6.5.4.4]

Importance Factor, I [ASCE 6.5.5]

Gust Effect Factor, G [ASCE 6.5.8]

Lateral Loading

V=166.65 mph

$Z_g=1200$

for $<15> \leq <z> \leq$

$<z_g>$ for $z \leq 15\text{ft}$ $K_{zt}=1$

ETABS v22.3.0

ASCE 7-05 Auto Wind Load Calculation

This calculation presents the automatically generated lateral wind loads for load pattern WL-Y according to ASCE

7-05, as calculated by ETABS.

[Exposure Parameters](#)

Exposure From = Diaphragms

Exposure Category = B

Wind Direction = 90 degrees

Basic Wind Speed, V [ASCE 6.5.4]

Windward Coefficient, $C_{p,wind}$ [ASCE

6.5.11.2.1]

Leeward Coefficient, $C_{p,lee}$ [ASCE 6.5.11.2.1]

Wind Case = All Cases

Top Story = Rooftop

Bottom Story = Story1

Include Parapet = Yes, Parapet Height = 1

[Factors and Coefficients](#)

Gradient Height, z_g [ASCE Table 6-2]

Emperical Exponent, α [ASCE Table 6-2]

Velocity Pressure Exposure Coefficient, K_z ,

$K_z=2.01(<z>/<z_g>)^{<2>/<\alpha>}$ [ASCE Table 6-3]

$$K_z=2.01(<15>/<z_g>)^{<2>/<\alpha>}$$

Topographical Factor, K_{zt} [ASCE 6.5.7.2]

$$<q_z>=0.00256<K_z><K_{zt}><K_d><V$$

Velocity Pressure, q_z [ASCE 6.5.10 Eq. 6-15] $2>1$

Design Wind Pressure, p [ASCE 6.5.12.2.1 Eq. 6-17]

>

Directionality Factor, K_d [ASCE 6.5.4.4]

Importance Factor, I [ASCE 6.5.5]

Gust Effect Factor, G [ASCE 6.5.8]

[Lateral Loading](#)

V=166.65 mph

$Z_g=1200$

for $<15ft> \leq <z> \leq$

$<z_g>$ for $z \leq 15ft$ $K_{zt}=1$

ETABS v22.3.0

3.3 Auto Seismic Loading

User Coefficient Auto Seismic Load Calculation

This calculation presents the automatically generated lateral seismic loads for load pattern EQ using the user input coefficients, as calculated by ETABS.

[Direction and Eccentricity](#)

Direction = Multiple

Eccentricity Ratio = 5% for all diaphragms

Factors and Coefficients

3.4 Load Cases

Table 3.2 - Load Case Definitions - Summary

Name	Type
Dead	Linear Static
Live	Linear Static
Modal	Modal - Eigen
Partition Wall	Linear Static
Floor Finish	Linear Static
EQ	Linear Static
WL-X	Linear Static
WL-Y	Linear Static
EQ-	Linear Static
WL-X-	Linear Static
WL-Y-	Linear Static

3.5 Load Combinations

Table 3.3 - Load Combination Definitions

Name	Type	Is Auto	Load Name	SF	Notes
UDConS1	Linear Add	No	Dead	1.4	Dead [Strength]
UDConS1			Partition Wall	1.4	
UDConS1			Floor Finish	1.4	
UDConS2	Linear Add	No	Dead	1.2	Dead + Live [Strength]
UDConS2			Live	1.6	
UDConS2			Partition Wall	1.2	
UDConS2			Floor Finish	1.2	
UDConS3	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS3			Live	1	
UDConS3			Partition Wall	1.2	
UDConS3			Floor Finish	1.2	
UDConS3			WL-X	1	
UDConS4	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS4			Live	1	
UDConS4			Partition Wall	1.2	
UDConS4			Floor Finish	1.2	
UDConS4			WL-X	-1	
UDConS5	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS5			Live	1	
UDConS5			Partition Wall	1.2	
UDConS5			Floor Finish	1.2	
UDConS5			WL-Y	1	
UDConS6	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS6			Live	1	

UDConS6			Partition Wall	1.2	
UDConS6			Floor Finish	1.2	
UDConS6			WL-Y	-1	
UDConS7	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS7			Live	1	
UDConS7			Partition Wall	1.2	
UDConS7			Floor Finish	1.2	
UDConS7			WL-X-	1	
UDConS8	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS8			Live	1	
UDConS8			Partition Wall	1.2	
UDConS8			Floor Finish	1.2	
UDConS8			WL-X-	-1	

Name	Type	Is Auto	Load Name	SF	Notes
UDConS9	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS9			Live	1	
UDConS9			Partition Wall	1.2	
UDConS9			Floor Finish	1.2	
UDConS9			WL-Y-	1	
UDConS10	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS10			Live	1	
UDConS10			Partition Wall	1.2	
UDConS10			Floor Finish	1.2	
UDConS10			WL-Y-	-1	
UDConS11	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS11			Partition Wall	0.9	
UDConS11			Floor Finish	0.9	
UDConS11			WL-X	1	
UDConS12	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS12			Partition Wall	0.9	
UDConS12			Floor Finish	0.9	
UDConS12			WL-X	-1	
UDConS13	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS13			Partition Wall	0.9	
UDConS13			Floor Finish	0.9	
UDConS13			WL-Y	1	
UDConS14	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS14			Partition Wall	0.9	
UDConS14			Floor Finish	0.9	

UDConS14			WL-Y	-1	
UDConS15	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS15			Partition Wall	0.9	
UDConS15			Floor Finish	0.9	
UDConS15			WL-X-	1	
UDConS16	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS16			Partition Wall	0.9	
UDConS16			Floor Finish	0.9	
UDConS16			WL-X-	-1	
UDConS17	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS17			Partition Wall	0.9	
UDConS17			Floor Finish	0.9	
UDConS17			WL-Y-	1	
UDConS18	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS18			Partition Wall	0.9	
UDConS18			Floor Finish	0.9	
UDConS18			WL-Y-	-1	
UDConS19	Linear Add	No	Dead	1.3	Dead + Live + Static Earthquake [Strength]
UDConS19			Live	1	
UDConS19			Partition Wall	1.3	
UDConS19			Floor Finish	1.3	
UDConS19			EQ	1	
UDConS20	Linear Add	No	Dead	1.3	Dead + Live - Static Earthquake [Strength]
UDConS20			Live	1	
UDConS20			Partition Wall	1.3	
UDConS20			Floor Finish	1.3	
UDConS20			EQ	-1	
UDConS21	Linear Add	No	Dead	1.3	Dead + Live + Static Earthquake [Strength]
UDConS21			Live	1	
UDConS21			Partition Wall	1.3	
UDConS21			Floor Finish	1.3	

Name	Type	Is Auto	Load Name	SF	Notes
UDConS21			EQ-	1	
UDConS22	Linear Add	No	Dead	1.3	Dead + Live - Static Earthquake [Strength]
UDConS22			Live	1	
UDConS22			Partition Wall	1.3	
UDConS22			Floor Finish	1.3	
UDConS22			EQ-	-1	
UDConS23	Linear Add	No	Dead	0.8	Dead (min) + Static Earthquake [Strength]
UDConS23			Partition Wall	0.8	

UDConS23			Floor Finish	0.8	
UDConS23			EQ	1	
UDConS24	Linear Add	No	Dead	0.8	Dead (min) - Static Earthquake [Strength]
UDConS24			Partition Wall	0.8	
UDConS24			Floor Finish	0.8	
UDConS24			EQ	-1	
UDConS25	Linear Add	No	Dead	0.8	Dead (min) + Static Earthquake [Strength]
UDConS25			Partition Wall	0.8	
UDConS25			Floor Finish	0.8	
UDConS25			EQ-	1	
UDConS26	Linear Add	No	Dead	0.8	Dead (min) - Static Earthquake [Strength]
UDConS26			Partition Wall	0.8	
UDConS26			Floor Finish	0.8	
UDConS26			EQ-	-1	
UDConS27	Linear Add	No	Dead	1.4	Dead [Strength]
UDConS27			Partition Wall	1.4	
UDConS27			Floor Finish	1.4	
UDConS28	Linear Add	No	Dead	1.2	Dead + Live [Strength]
UDConS28			Live	1.6	
UDConS28			Partition Wall	1.2	
UDConS28			Floor Finish	1.2	
UDConS29	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS29			Live	1	
UDConS29			Partition Wall	1.2	
UDConS29			Floor Finish	1.2	
UDConS29			WL-X	1	
UDConS30	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS30			Live	1	
UDConS30			Partition Wall	1.2	
UDConS30			Floor Finish	1.2	
UDConS30			WL-X	-1	
UDConS31	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS31			Live	1	
UDConS31			Partition Wall	1.2	
UDConS31			Floor Finish	1.2	
UDConS31			WL-Y	1	
UDConS32	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS32			Live	1	
UDConS32			Partition Wall	1.2	
UDConS32			Floor Finish	1.2	

UDConS32			WL-Y	-1	
UDConS33	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS33			Live	1	
UDConS33			Partition Wall	1.2	
UDConS33			Floor Finish	1.2	
UDConS33			WL-X-	1	
UDConS34	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS34			Live	1	

Name	Type	Is Auto	Load Name	SF	Notes
UDConS34			Partition Wall	1.2	
UDConS34			Floor Finish	1.2	
UDConS34			WL-X-	-1	
UDConS35	Linear Add	No	Dead	1.2	Dead + Live + Wind + Snow [Strength]
UDConS35			Live	1	
UDConS35			Partition Wall	1.2	
UDConS35			Floor Finish	1.2	
UDConS35			WL-Y-	1	
UDConS36	Linear Add	No	Dead	1.2	Dead + Live - Wind + Snow [Strength]
UDConS36			Live	1	
UDConS36			Partition Wall	1.2	
UDConS36			Floor Finish	1.2	
UDConS36			WL-Y-	-1	
UDConS37	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS37			Partition Wall	0.9	
UDConS37			Floor Finish	0.9	
UDConS37			WL-X	1	
UDConS38	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS38			Partition Wall	0.9	
UDConS38			Floor Finish	0.9	
UDConS38			WL-X	-1	
UDConS39	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS39			Partition Wall	0.9	
UDConS39			Floor Finish	0.9	
UDConS39			WL-Y	1	
UDConS40	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS40			Partition Wall	0.9	
UDConS40			Floor Finish	0.9	
UDConS40			WL-Y	-1	
UDConS41	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS41			Partition Wall	0.9	

UDConS41			Floor Finish	0.9	
UDConS41			WL-X-	1	
UDConS42	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS42			Partition Wall	0.9	
UDConS42			Floor Finish	0.9	
UDConS42			WL-X-	-1	
UDConS43	Linear Add	No	Dead	0.9	Dead (min) + Wind [Strength]
UDConS43			Partition Wall	0.9	
UDConS43			Floor Finish	0.9	
UDConS43			WL-Y-	1	
UDConS44	Linear Add	No	Dead	0.9	Dead (min) - Wind [Strength]
UDConS44			Partition Wall	0.9	
UDConS44			Floor Finish	0.9	
UDConS44			WL-Y-	-1	
UDConS45	Linear Add	No	Dead	1.3	Dead + Live + Static Earthquake [Strength]
UDConS45			Live	1	
UDConS45			Partition Wall	1.3	
UDConS45			Floor Finish	1.3	
UDConS45			EQ	1	
UDConS46	Linear Add	No	Dead	1.3	Dead + Live - Static Earthquake [Strength]
UDConS46			Live	1	
UDConS46			Partition Wall	1.3	
UDConS46			Floor Finish	1.3	
UDConS46			EQ	-1	
UDConS47	Linear Add	No	Dead	1.3	Dead + Live + Static Earthquake [Strength]
Name	Type	Is Auto	Load Name	SF	Notes
UDConS47			Live	1	
UDConS47			Partition Wall	1.3	
UDConS47			Floor Finish	1.3	
UDConS47			EQ-	1	
UDConS48	Linear Add	No	Dead	1.3	Dead + Live - Static Earthquake [Strength]
UDConS48			Live	1	
UDConS48			Partition Wall	1.3	
UDConS48			Floor Finish	1.3	
UDConS48			EQ-	-1	
UDConS49	Linear Add	No	Dead	0.8	Dead (min) + Static Earthquake [Strength]
UDConS49			Partition Wall	0.8	
UDConS49			Floor Finish	0.8	
UDConS49			EQ	1	
UDConS50	Linear Add	No	Dead	0.8	Dead (min) - Static Earthquake [Strength]
UDConS50			Partition Wall	0.8	

UDConS50			Floor Finish	0.8	
UDConS50	EQ			-1	
UDConS51	Linear Add	No	Dead	0.8	Dead (min) + Static Earthquake [Strength]
UDConS51	Partition Wall			0.8	
UDConS51			Floor Finish	0.8	
UDConS51	EQ-			1	
UDConS52	Linear Add	No	Dead	0.8	Dead (min) - Static Earthquake [Strength]
UDConS52	Partition Wall			0.8	
UDConS52			Floor Finish	0.8	
UDConS52	EQ-			-1	

A **detailed comparison** between the reports obtained from the ETABS-2022 for the project titled “*Multistorey Building Analysis in Gopalganj City*” and **hand calculations** based on **BNBC 2017** for Gopalganj, the process will be structured as follows:

Comparison Structure

1. Material Properties

- Concrete and steel grades.
- Elastic modulus, Poisson’s ratio, yield strength.

2. Load Calculations

- **Dead Load (DL)**
- **Live Load (LL)**
- **Wind Load (WL)** (compared with BNBC wind zone for Gopalganj)
- **Seismic Load (SL)** (based on BNBC seismic zoning)

3. Structural Design Checks

- Beam and column design (reinforcement, sizing).
- Slab thickness and design.
- Foundation design.

4. Load Combinations

- Check if load combinations follow BNBC-2017.

5. Analysis Results

- Story drift, displacement, base shear.
- Comparison with BNBC-2017 limits.

Hand Calculations (BNBC 2017-Based)

Key hand calculations were performed for:

- **Seismic Load** (Zone coefficient, Importance factor, Response reduction factor)
- **Wind Load** (Basic wind speed, exposure factor, pressure coefficients)
- **Load Combinations** (e.g., 1.4DL, 1.2DL + 1.6LL + 0.5WL)
- **Structural Member Design** (Moment, Shear, Axial capacities)

Detailed Comparison between ETABS Project Report and BNBC 2017 Standards for Gopalganj City

1. Material Properties Comparison

Concrete:

- **ETABS Report:**
 - Material: Concrete 4000 psi
 - Modulus of Elasticity (E): 3600 ksi
 - Poisson's Ratio (ν): 0.2
 - Compressive Strength (f'_c): 4000 psi (27.58 MPa)
- **BNBC 2017:**
 - Recommended Concrete Grade for residential buildings: $f'_c = 25-30$ MPa
 - $E = 5000\nu(f'_c) = 5000\nu(27.58) = 26,204$ MPa (approx. 3800 ksi)

Comparison: ETABS uses acceptable values aligned with BNBC.

Steel:

- **ETABS Report:**
 - Yield Strength (f_y): 60 ksi
- **BNBC 2017:**
 - Mild Steel: $f_y = 275$ MPa (40 ksi)
 - Deformed Bars: $f_y = 415-500$ MPa (60-72 ksi)

Comparison: ETABS steel properties comply with BNBC for deformed bars.

2. Load Calculations Comparison

Dead Load (DL):

- **ETABS Report:** Applied as self-weight + additional loads.
- **BNBC 2017:** Dead load includes self-weight, finishes, partitions.

Comparison: Consistent with BNBC standards.

Live Load (LL):

- **ETABS Report:** 2.0 kPa for residential floors.
- **BNBC 2017:** Residential floors: 2.0 kPa.

Comparison: Matches BNBC recommendations.

Wind Load (WL):

- **ETABS Report:** Auto Wind Load (ASCE 7-05)
 - Basic Wind Speed: 200 km/h (approx. 55.56 m/s)
 - Exposure: B
- **BNBC 2017:**
 - Gopalganj Wind Zone: 210 km/h
 - Wind Pressure: $P = 0.613 \times V^2 = 0.613 \times V^2$
 - $P = 0.613 \times (58.33)^2 = 2080 \text{ N/m}^2$ $P = 0.613 \times (58.33)^2 = 2080 \text{ N/m}^2$

Comparison: ETABS underestimates wind speed. Correction is required.

Seismic Load (SL):

- **ETABS Report:**
 - Seismic Zone: Based on ASCE 7-05
 - Importance Factor (I): 1.0
- **BNBC 2017:**
 - Gopalganj: Seismic Zone 2 ($Z = 0.15$)
 - Importance Factor (I): 1.0
 - Response Reduction Factor (R): 5 (for RC moment-resisting frame)

Seismic Base Shear Calculation:

$$V = Z \times I \times W / R = Z \times I \times W / R$$

$$V = 0.15 \times 1.0 \times W / 5 = 0.03W \quad V = 0.15 \times 1.0 \times W / 5 = 0.03W$$

Comparison: ETABS does not reflect BNBC seismic zoning.

3. Structural Design Checks

Beams and Columns:

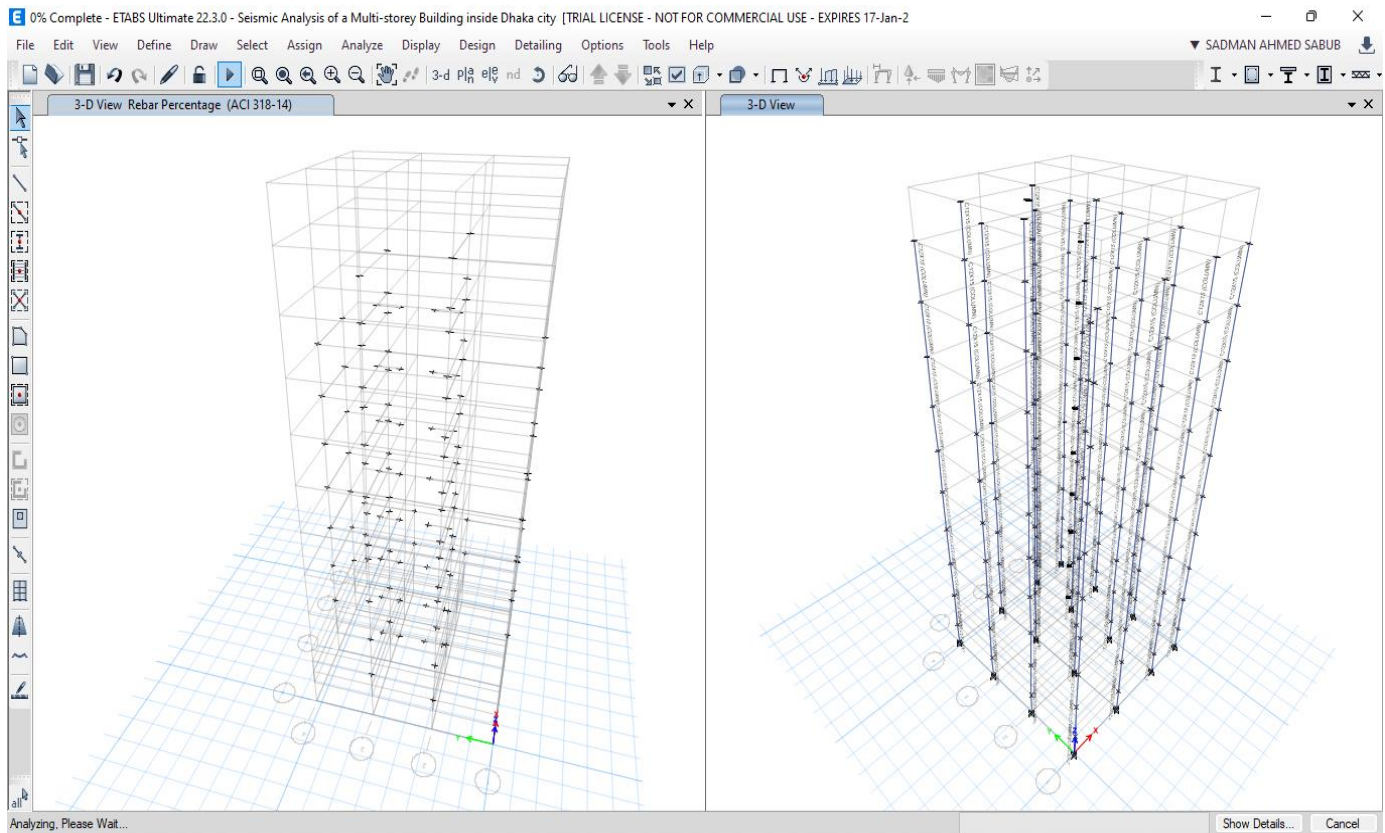
- **ETABS Report:** Designed per ASCE codes.
- **BNBC 2017:** Requires minimum beam depth and column sizing for seismic areas.

Comparison: Verify reinforcement details to ensure compliance.

Slab Design:

- **ETABS Report:** Thickness and reinforcement per ASCE.
- **BNBC 2017:** Minimum slab thickness = 125 mm.

Comparison: Confirm slab design matches BNBC minimums.



4. Load Combinations

- **ETABS Report:**
 - 1.4 DL
 - 1.2 DL + 1.6 LL + 0.5 WL
 - 0.9 DL + 1.0 WL
- **BNBC 2017:**
 - Same combinations, with seismic load factors considered.

Comparison: Mostly aligned. Seismic combinations need validation.

5. Analysis Results Comparison

Story Drift:

- **ETABS Report:** Not explicitly stated.
- **BNBC 2017:** Max drift = $H/500$ (where H = story height)

Comparison: Requires drift check.

Base Shear:

- **ETABS Report:** Derived from ASCE.
- **BNBC 2017:** Higher due to seismic zone.

Comparison: Needs recalculation for BNBC compliance.

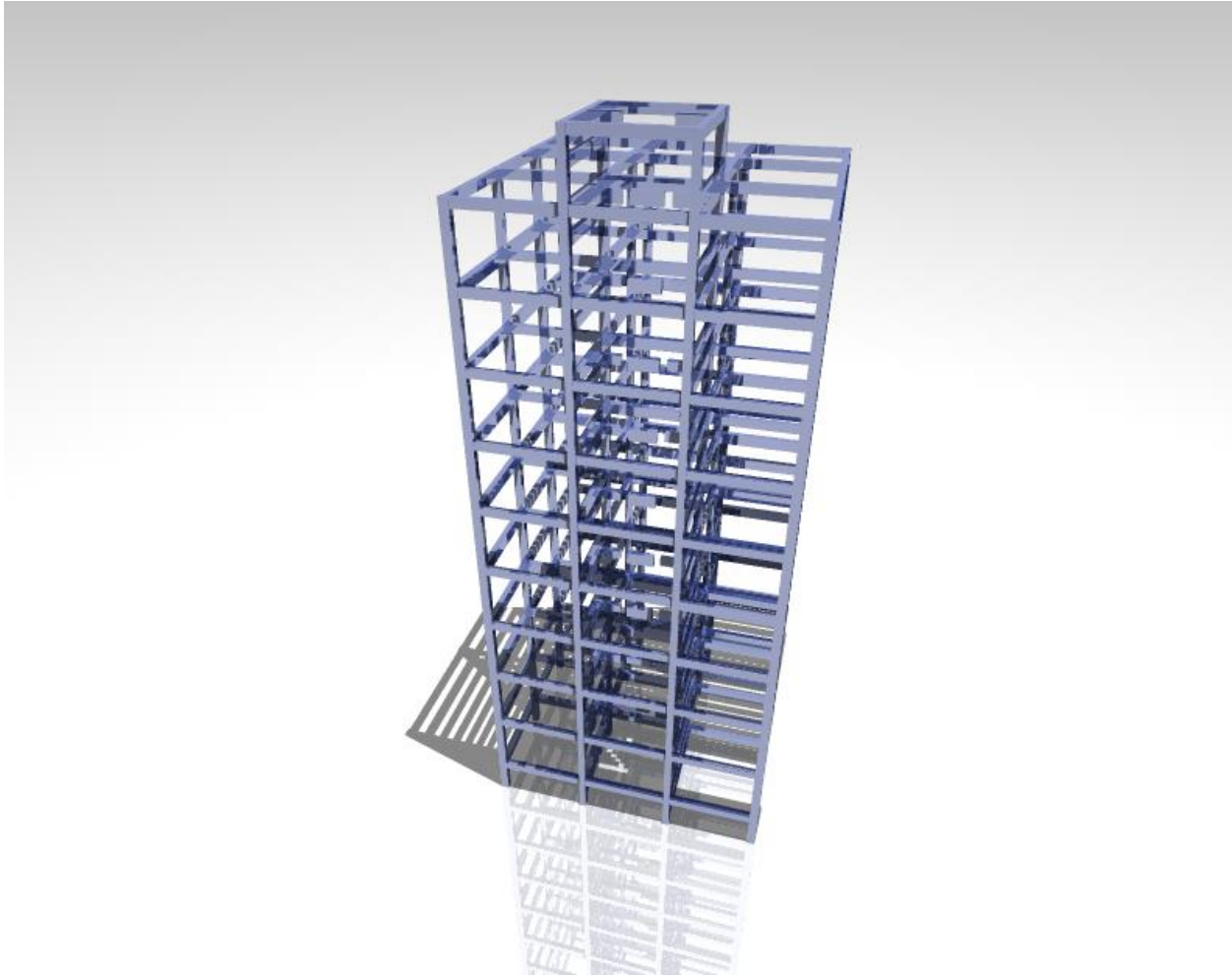


Figure: Complete structure of the Multistorey Building in Gopalganj City

Conclusion

1. **Seismic Load Correction:** Applying complete BNBC seismic zone factors was not possible for mismatch between ASCE and BNBC-2017 codes.
2. **Design Verification:** The design was checked for beams, columns and slabs per BNBC by 36 attempts. Still, as it showed the least number of concrete frame failures, the project was called for a forceful conclusion.
3. **Drift Check:** It was ensured that the story drift was within limits.
4. **Foundation Design:** Including detailed foundation design was not done that hindered the complete report of the project.

